

CHIRAG APARADH

Software Engineer | Backend • Distributed Systems • AI

@ chiragaparadh@gmail.com

✉ Mumbai, India

🔗 git-chirag

📞 +919833039704

🌐 chirag-aparadh

🌐 Portfolio

EXPERIENCE

Software Engineer (Associate Consultant)

Oracle Financial Services Software

📅 Jul 2023 – Present

📍 Mumbai, MH

- Built and maintained core **Java EE microservices** powering HDFC Bank's net banking platform (**2M+ daily active users**), delivering **99.99% up-time** through highly available clustered deployments.
- Implemented a **bot detection workflow** using **BioCatch** for merchant transactions (e.g., Amazon, Flipkart), triggering **OTP-based authentication** when suspicious activity was flagged, strengthening fraud prevention for **150K daily transactions**.
- Built event-driven integrations using **Apache Kafka** to publish acknowledgment messages to external broker systems (e.g., Zerodha) after successful fund transfers, enabling **reliable and decoupled communication** across banking services.
- Modernized a legacy banking workflow by translating UI-driven business logic into **backend service APIs**, performing **end-to-end debugging** across service and database layers and improving reliability through **standardized data contracts** and validation.
- Technologies Used** : Java, JavaScript, SQL, Oracle Database, Apache Kafka, Microservices, IBM WebSphere

Machine Learning Engineer - Intern

TechCiti

📅 Aug 2021 – Sep 2021

📍 Bangalore - Remote, KA

- Designed a website to **assess loan eligibility** of an individual by analyzing 5 crucial parameters: **age, loan amount, credit score, location, and employment type**.
- Developed and trained a logistic regression model achieving **92% prediction accuracy** on test data.
- The website consisted of 3 key pages: **Sign up/Login, Profile updation page, and eligibility prediction page**.
- Technologies Used** : Python, Numpy, Pandas, scikit-learn

EDUCATION

M.S. in Computer Science

University of Massachusetts Amherst

📅 Sep 2026 – May 2028 (Expected)

📍 Amherst, MA, USA

B.Tech in Computer Science & Engineering

Ramrao Adik Institute of Technology

📅 Mumbai University, 2023

📍 CGPA: 9.51 / 10

ACHIEVEMENTS

- Finalist in the **Meta × Hugging Face OpenEnv Hackathon**, advancing to the national finale among **31,000+ registered teams**.
- Ranked **1108** globally in **LeetCode Biweekly Contest 145** and earned the **Knight** badge.
- Achieved **5-Star Gold Badge** in **Problem Solving** on HackerRank.
- Cleared **GATE CS 2023** with an **AIR 2816** among **75,680** candidates.

INTERESTS

- Competitive Programming, Chess, Weight Training, AI Systems

TECHNICAL SKILLS

- Languages**: Java, Python, C, JavaScript, SQL
- Backend & Technologies**: Java EE, FastAPI, Django, React, REST APIs, Microservices, Apache Kafka, Redis
- Databases & Cloud**: Oracle Database, MySQL, MongoDB, Fire-base, Qdrant, Amazon S3, AWS, Docker, Kubernetes, Amazon ECS, Amazon ECR, Jenkins, Linux
- AI & Core CS**: PyTorch, CLIP, NumPy, Pandas, Scikit-learn, DSA, OOP, Operating Systems, Computer Networks, Distributed Systems

CODING PROFILE

LeetCode - eLeet_chirag | Rating: 1866

HackerRank - chiragaparadh

PROJECTS

Multimodal Image Search System

- Developed a FastAPI-based asynchronous image processing service generating **512-dimensional CLIP embeddings** for semantic image understanding.
- Indexed image vectors in **Qdrant** to support text-to-image retrieval, visual similarity search, and near-duplicate detection.
- Designed a distributed image processing pipeline using **Celery** and **Redis** for asynchronous task execution and real-time status tracking.
- Containerized and deployed the application using **Docker, AWS ECR, and Amazon ECS**.
- Technologies Used**: FastAPI, Python, PyTorch, CLIP, Qdrant, Redis, Celery, Amazon S3, Docker, AWS ECS, AWS ECR

Supply Chain Management Using Blockchain

- Developed a **decentralized application using Blockchain** technology, specifically Ethereum for the Blockchain Network and React for the user interface.
- The application **improves transparency and traceability** of products and pricing within the supply chain and has 4 distinct roles: **Farmer/Producer, Distributor, Retailer, and Consumer**.
- Implemented a comprehensive set of 13 functions through smart contracts to effectively oversee various supply chain activities, including **product listing, bidding, selling, and purchasing**.
- Technologies Used** : Ethereum, Solidity, React, Tailwind

Social Meetup Application

- Created a Flutter application that **assists people in planning their meetups**.
- The application allows people to send connection requests and **view each other's locations on a map**.
- The app **suggests nearby restaurants, cafes, and malls** that are convenient for all friends, with directions to reach these places.
- Technologies Used** : Flutter, Dart, Firebase, Google Maps API

PUBLICATION

- Exploring the Role of Blockchain in Enhancing Supply Chain (IEEE, Sep '23)